Basic Electrical Engineering - Original Notes

Unit 1: DC Circuits

Ohm's law: $V=IR$. Kirchhoff's Current Law: sum of currents at a node is zero. Kirchhoff's Voltage Law: sum of voltages in a loop is zero. Series and parallel resistor analysis.

Unit 2: AC Fundamentals

Sinusoidal voltage/current, RMS value, average value, frequency, phase angle, impedance, power factor.

Unit 3: Magnetic Circuits & Transformers

Magnetic flux, MMF, reluctance, Faraday's law, transformer construction, EMF equation, losses and efficiency.

Unit 4: Electrical Machines

Overview of DC machines, induction motors and synchronous machines, applications and basic operating principles.

Unit 5: Power & Safety

Active, reactive and apparent power, earthing, electrical safety, protective devices such as MCBs and fuses.

Important Formulas

$V=IR$, $P=VI$, $P=I^2R$, $P=V^2/R$, $XL=2\pi fL$, $XC=1/(2\pi fC)$, $Z=\sqrt{(R^2+X^2)}$, $PF=\cos\varphi$.

Exam Tips

Practice circuit problems, learn definitions, draw neat diagrams, memorize key formulas.